

KARTHIK MANNEM

Los Angeles, CA

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SUMMARY

Machine Learning Engineer who approaches problem-solving as an end-to-end AI pipeline, ingesting complex real-world inputs, performing structured feature engineering, training and fine-tuning deep learning and transformer-based architectures, and deploying scalable inference systems. With 3+ years of experience across predictive modeling, computer vision, and LLM-powered Retrieval-Augmented Generation (RAG) workflows, I focus on evaluation rigor, performance optimization, and production-grade deployment.

PERSONAL PROJECTS

•Retail Demand Intelligence Platform – Forecasting, Anomaly Detection & Agentic Q&A

End-to-end forecasting system with a tool-calling LLM agent and a full AWS deployment.

- Built a 3-model demand forecasting pipeline (seasonal-naive, Holt-Winters, gradient-boosted) across 240 store/department series with rolling-origin backtesting, selecting the winning model per series (gradient-boosted: 7.9% WAPE, won 154/240 series); shipped 80% prediction intervals and occlusion-based feature attribution, validating both against held-out data (74.2% measured interval coverage vs. 80% nominal) rather than asserting calibration.
- Designed a 6-tool agentic Q&A layer with a pluggable LLM backend (Anthropic/OpenAI/Bedrock), full audit-trail logging, retry/backoff with token-cost tracking, and a 37-case labeled eval harness scoring tool-routing accuracy in CI.
- Closed the MLOps loop end-to-end: automated drift detection triggers per-series model re-selection with a logged audit trail, deployed to AWS (SageMaker, Lambda, Bedrock) via Terraform with CI and Docker verified passing.
- **Tech:** Python, scikit-learn, Flask, Anthropic/OpenAI/Bedrock APIs, Terraform, AWS (SageMaker, Lambda, API Gateway), Docker, GitHub Actions

•Preventive Care Recommendation Agent | LLM + RAG (LangChain)

LLM-powered healthcare decision support system deployed publicly.

- Designed and deployed an LLM-powered preventive care recommendation system that analyzes patient age, medical history, and risk factors to generate personalized screening and vaccination guidance aligned with CDC guidelines.
- Architected a Retrieval-Augmented Generation (RAG) pipeline using LangChain, implementing document chunking, embeddings, and vector-based semantic retrieval.
- Applied prompt engineering techniques to improve reasoning accuracy and reduce hallucinated medical outputs.
- **Tech:** Python, LangChain, OpenAI API, Vector Search, Flask, Docker

•SplitWise Pro – AI-Powered Expense Sharing Platform

Full-stack expense management system with OCR-based receipt parsing and cloud deployment.

- Built a full-stack expense sharing platform for group expense tracking, receipt uploads, balance calculation, and optimized settlement generation.
- Integrated Firebase Authentication with a Node.js/Express backend to secure protected routes and synchronize authenticated users with MongoDB Atlas.
- Implemented AWS S3-based receipt storage with Google Vision API and OpenAI-assisted OCR parsing to extract structured expense details.
- **Tech:** Next.js, TypeScript, Node.js, Express, MongoDB Atlas, Firebase Auth, AWS S3, Google Vision API, OpenAI API

•Estimating Product Length using BERT on Amazon Dataset

NLP-based regression model for structured e-commerce prediction.

- Fine-tuned a pre-trained BERT model using product titles, descriptions, and category metadata to predict product length in a large-scale e-commerce dataset.
- Implemented text preprocessing, tokenization, and feature engineering pipelines to improve model convergence and prediction stability.
- Applied log transformation and model ensembling techniques to enhance regression performance and reduce prediction variance.
- **Tech:** Python, BERT, HuggingFace Transformers, Scikit-learn, Pandas

TECHNICAL SKILLS

Programming: Python, Java
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV
NLP & LLM Systems: Transformers, HuggingFace, LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG)
Data Processing: Pandas, NumPy, Feature Engineering, Data Preprocessing
Deployment & Tools: Docker, Flask, REST APIs, Git, GitHub
Databases: MongoDB, PostgreSQL, MySQL
Cloud: AWS (EC2, S3)

EDUCATION

- California State University, Northridge

Master of Science in Computer Science | **CGPA: 3.92 / 4.0**

Los Angeles, CA
Aug 2024 – May 2026
- Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering | **CGPA: 9.21 / 10.00**

Chennai, India
Aug 2020 – May 2024

EXPERIENCE

- AI Model Evaluation Analyst

Handshake AI

Sep 2025 – April 2026

- Evaluated 1,000+ large language model (LLM) outputs using structured quality benchmarks to assess response accuracy, coherence, and factual consistency.
 - Identified recurring model failure cases and categorized hallucination patterns, contributing to improved prompt refinement workflows and reducing evaluation inconsistencies by 20%.
 - Collaborated with cross-functional teams to provide systematic feedback for improving model behavior across multiple domain-specific evaluation tasks.
 - Maintained high annotation agreement standards across large datasets, increasing benchmarking reliability and validation consistency.
- Machine Learning Engineer (Freelance)

Self-Employed

Aug 2023 – June 2024

- Designed and deployed deep learning models (LSTM, CNN) for 3+ client applications, delivering production-ready inference pipelines.
 - Built end-to-end ML workflows including data preprocessing, feature engineering, model training, and cross-validation, improving model training efficiency by 30%.
 - Containerized ML services using Docker and exposed REST APIs via Flask, enabling scalable real-time prediction systems with reduced deployment overhead.
 - Optimized data pipelines and training configurations, reducing overall training time by 25% and improving inference latency in deployed systems.

POSITIONS OF RESPONSIBILITY

- Board of Directors – Facilities and Operations Specialist

University Student Union, CSUN

Aug 2024 – May 2026

- Oversaw campus infrastructure initiatives impacting 38,000+ students and managed a \$5M operational budget.
 - Collaborated with executive leadership to improve sustainability, accessibility, and space utilization across university facilities.
- Content Lead

Google Developer Student Clubs

Aug 2022 – Apr 2024

- Led a team of 8+ members, organizing workshops and publishing technical content for 200+ participants.
 - Mentored contributors in Git/GitHub workflows and coordinated cross-campus technical collaborations.

LICENSES AND CERTIFICATIONS

Artificial Intelligence and Machine Learning (Foundation and Associate) – AWS
Fundamentals of Deep Learning – NVIDIA
Program Essentials in Python Programming (Intermediate Level) – CISCO
Model Validation 2 Expert – Handshake AI